Designing 'Sortino', an automatic waste sorting system, with the implementation of Deep Learning for a University

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INTRODUCTION

This paper presents Sortino, a Raspberry Pi-based intelligent waste sorting system designed for university settings. Leveraging deep learning and IoT, the system uses a YOLOv8l model for real-time classification of biodegradable, plastic, and metal waste via a camera module. Integrated with a Blazor-based web application and TensorFlow, Sortino actuates servo motors to sort waste into corresponding bins. A REST API enables seamless communication between the front end and hardware components. The system features a gamified rewards mechanism via QR code scans to encourage user engagement. Functional and usability evaluations, including a System Usability Scale (SUS) test, showed high reliability and user satisfaction. Sortino demonstrates a scalable, practical approach to promoting sustainable waste practices in academic environments.

OBJECTIVES

This study aims to develop and implement Sortino, an automated waste sorting system that utilizes Raspberry Pi technology and servo-controlled lids for efficient waste segregation. It includes the creation of a web application to track user points and guide waste disposal, alongside the integration of a REST API to enable seamless communication between the web app and hardware components. The system's performance will be evaluated based on its classification accuracy using image recognition. Additionally, Sortino seeks to promote sustainable waste management practices within the university, encouraging environmentally responsible behavior among students and staff while contributing to broader sustainability goals.

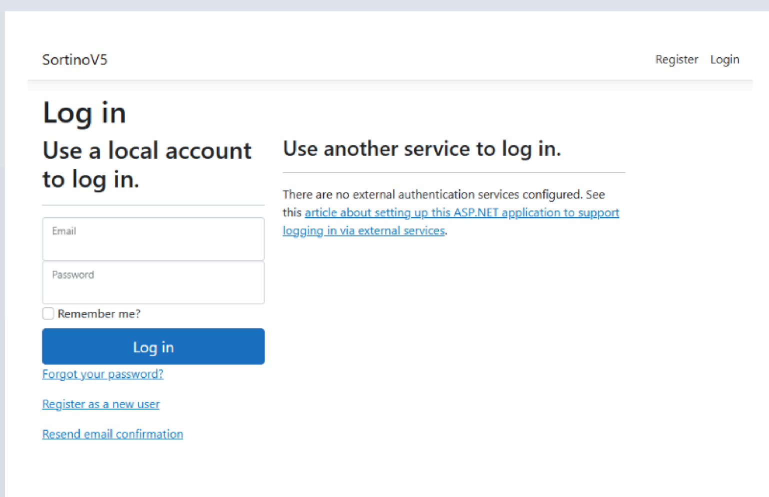
USER INTERFACE

The Sortino web application was designed with user-friendly simplicity to allow students and staff to efficiently interact with the smart bin system. Key features include user account creation, login, and profile management. The system incorporates a rewards mechanism where users earn points for correctly segregating waste. Each user is assigned a unique QR code linked to their Sortino web app account, which is scanned during waste disposal to ensure accurate point allocation. After scanning, users are redirected to a dedicated page where they can view, manage, and redeem their accumulated points.

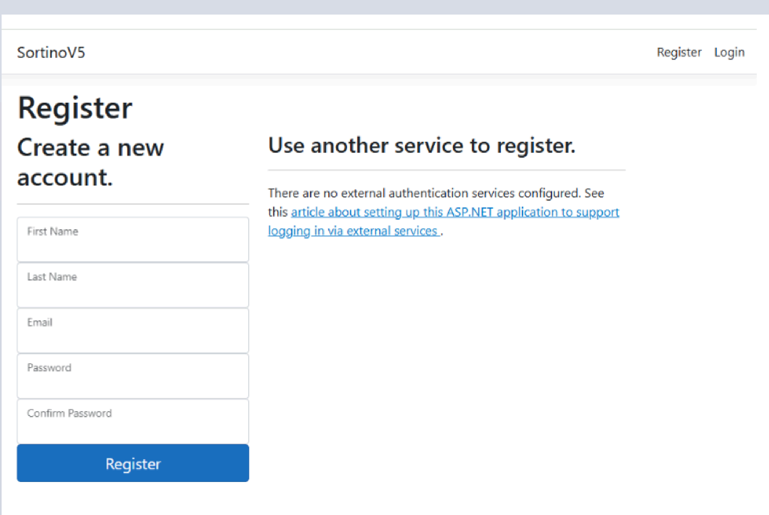
SORTINO

Your trusted smart bin!

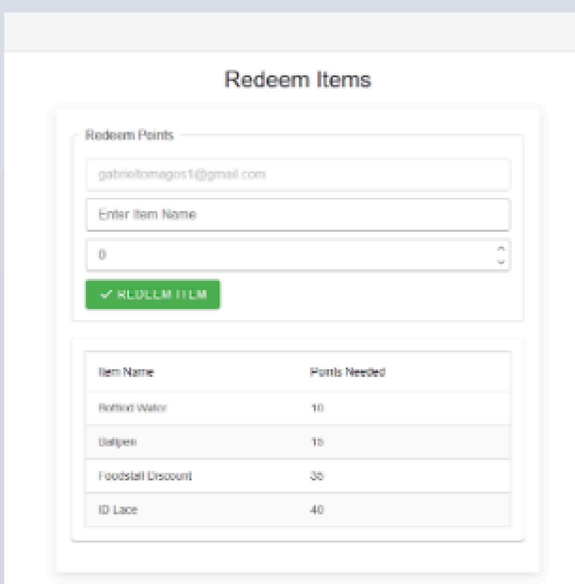
Official Logo



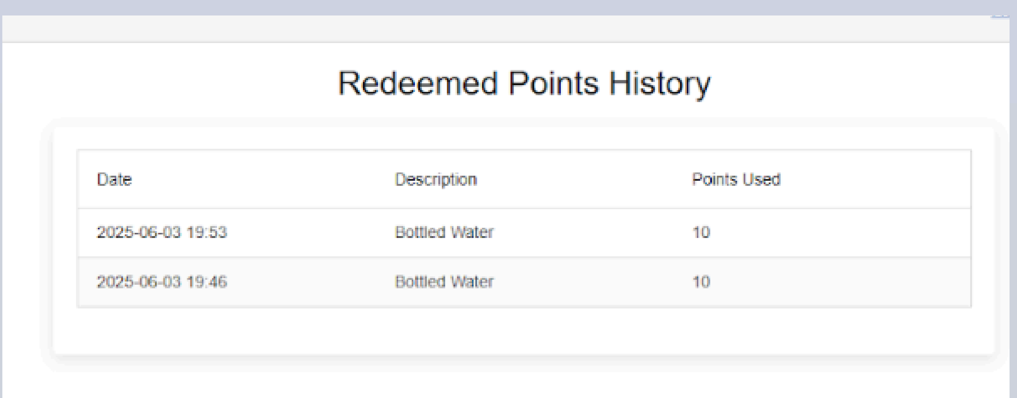
Log In Page



Registration page



Redeem Items Page



Redeemed Points History Page

Sample of generated QR

RESULTS

The usability of the Sortino system was assessed using the System Usability Scale (SUS), a widely accepted tool for evaluating user experience. A total of 20 respondents participated, including 15 students and five maintenance personnel, who rated the system on a five-point Likert scale. The student group reported an average SUS score of 80.17, while the maintenance staff recorded a score of 79.5. When combined, the system achieved an overall average score of 80.0. According to standard SUS interpretation guidelines, scores above 80 indicate "Good to Excellent" usability. These results suggest that the Sortino system is intuitive, user-friendly, and effectively supports its intended tasks across diverse user groups.

Although a formal User Acceptance Testing (UAT) phase was not conducted, the structure and outcomes of the user testing phase effectively served the same purpose. Participants could interact with the entire system—from disposing of waste in the smart bin to navigating the web interface and claiming rewards—providing a holistic view of its functionality. Positive feedback from both student users and maintenance personnel demonstrated strong acceptance. Users highlighted the system's ease of use, functional reliability, and the engagement fostered by the reward-based point system. Notably, even users with minimal technical background, such as some maintenance staff, could operate the system confidently, underscoring its inclusive design. While the system was generally well-received, participants suggested improvements, including enhancing the accuracy of biodegradable waste detection, reducing system latency, and refining the web interface layout. These insights indicate that the Sortino system meets the core expectations of its target users and is functionally ready for real-world application. However, further optimizations are recommended for broader deployment.

CONCLUSION

The Sortino system was developed to tackle waste management challenges in a university setting through an automated, deep learning–powered sorting mechanism. System testing and user evaluations indicate that Sortino effectively enhances waste sorting efficiency, promotes responsible disposal behaviours, and reduces environmental impact through its reward-based approach. The successful integration of the YOLOv8l model into a Blazor-based web application and REST API communication to the Raspberry Pi 5 enabled accurate and responsive waste classification. Usability testing confirmed strong user acceptance, though aspects such as QR code scanning could benefit further refinement. Overall, Sortino demonstrates significant potential as a scalable and sustainable solution for innovative waste management within academic institutions and similar environments.

The Sortino system has shown promising results in automating waste sorting within a university setting, but several enhancements are needed for improved performance, user engagement, and scalability. Key areas for development include improving the classification accuracy of biodegradable waste through dataset expansion and model fine-tuning, streamlining user identification with alternatives like NFC, optimising the web interface, and integrating a dedicated mobile app. The system should adopt cloud-based infrastructure for broader deployment, support higher traffic and diverse waste types, and incorporate IoT features such as weight sensors and bin-level indicators. Additionally, integrating digital payment systems for reward redemption and conducting long-term environmental impact assessments will be vital in refining Sortino into a scalable and sustainable innovative waste management solution.

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